ECE0202: Embedded Processors and Interfacing

Lab 1: Installation of KEIL Microcontroller Development Kit (MDK) and Project Creation

Objectives

- **Become familiar with running a project in KEIL**
- **Become familiar with the KEIL debugger interface**
- **Modify assembly code to store custom data into RAM**

Deliverables – total 50 points

- **(25 points) Modified code that saves a new hex code of your choice into memory.**
- **(25 points) A screenshot of your last names saved into memory from the debugger.**

Submit your code through Black board as a *.PDF file and the pictures as *.pdf documents.

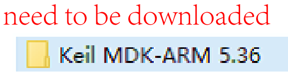
Section 1: Installation of Keil

Warning: Do not connect the lab Kit into your PC or laptop before the software installation completes. If you connect your kit to PC before installing the USB driver, Windows OS often mistakenly associates a wrong USB driver to the kit. Thus, you will not be able to program the kit. The solution is to go to the control panel and change the USB driver to ST-Link USB driver.

Step 1: Install Keil MDK-ARM

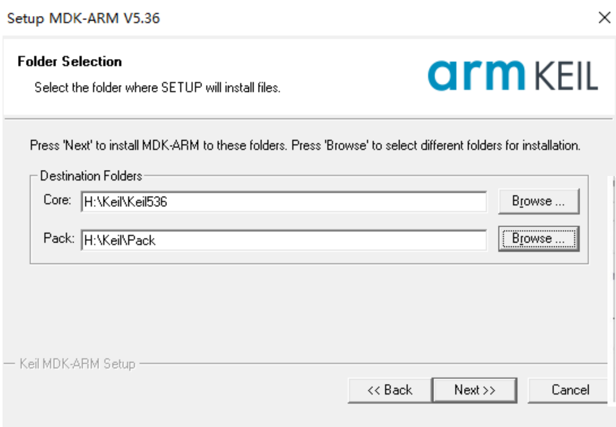
1. Download the latest free evaluation version Keil MDK-ARM from the following link:

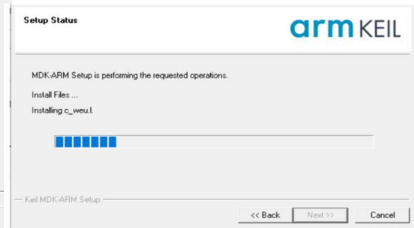
   https://www.keil.com/demo/eval/arm.htm.

   Version 5.36 Install pack is available at the lab and does not need to be downloaded from the website 

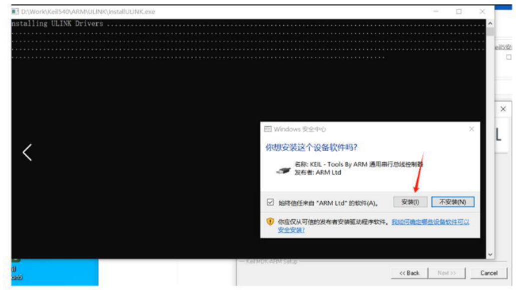
   - Keil MDK-ARM contains µVision 5 IDE (Integrated Development Environment) with debugger, flash programmer and the ARM compiler toolchain.
   - The major limitation of the free version is that programs that generate more than 32 Kbytes of code and data will not compile, assemble, or link.

2. Run the downloaded MDK5xx.exe . The software takes 2GB disk storage space. You should better install it to a different driver, instead of the default C drive.

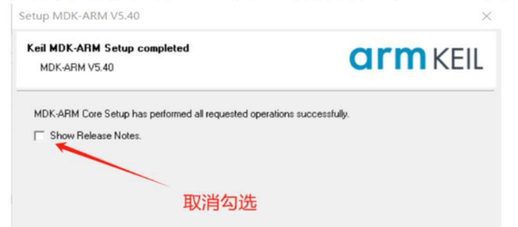




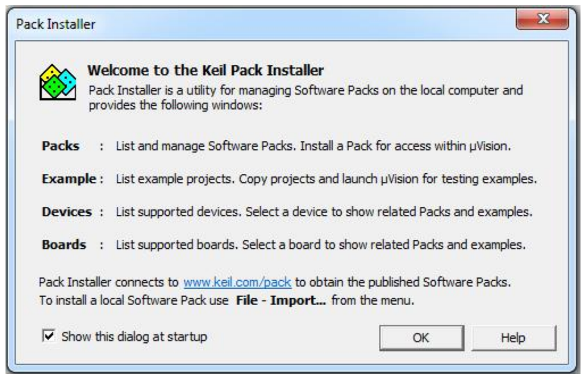
安装串行总线控制器，初次安装会弹，安装过 keil 可能不会有此弹窗



安装完成后会弹出如下窗口，取消勾选展示发行说明，然后点击 finish 完成。

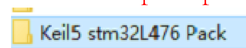


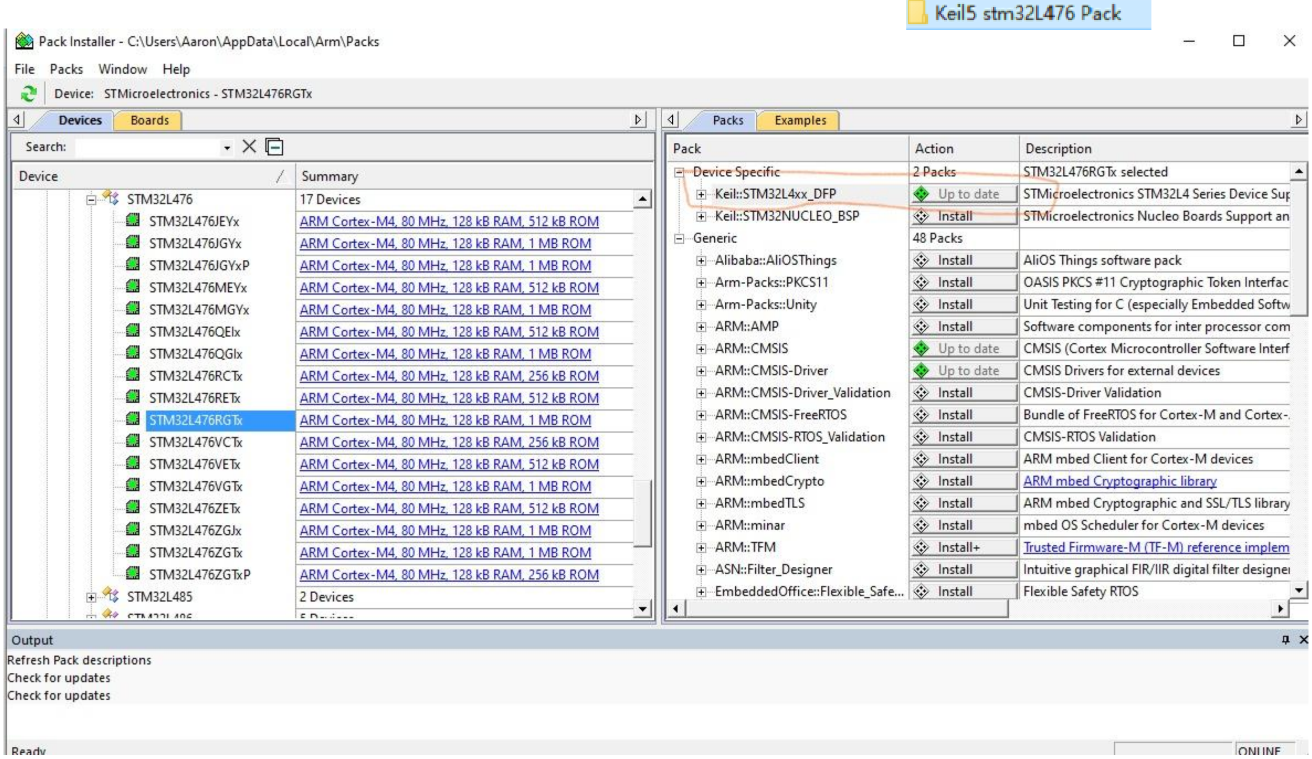
After the core software is installed, a dialog will show up to install Keil Pack. It automatically downloads selected components (called packs) from http://www.keil.com/dd2/pack/



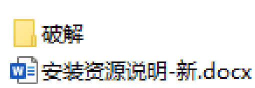
Click OK and then the following window shows up. Select the device **STM32L476RGTx**, install **STM32L4xx_DFP**.

Pack download is slow on this page, close this page and use the Install pack provided by the lab





After installing the stm32 L476RGTx pack, Use cracking software to crack, see "Installation Resource Notes - New"



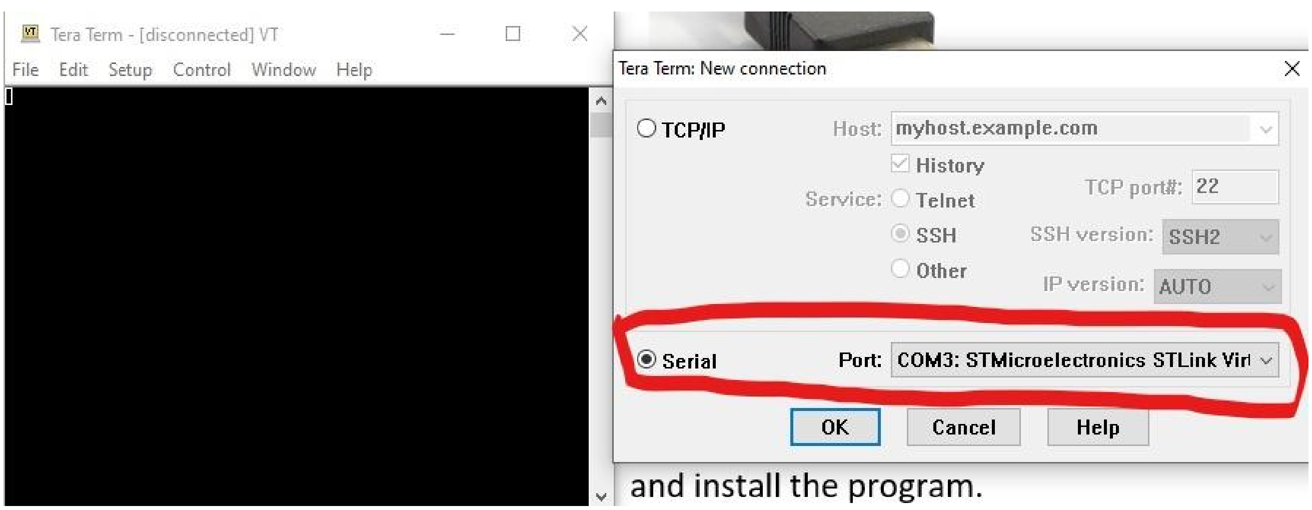
Step 2: Install ST-Link USB Driver

- Do not connect the lab kit before you install the USB driver for ST-Link.
- Go to the directory **C:\Keil_v5\ARM\STLink\USBDriver** and run **stlink_winusb_install.bat** in **administrator mode**.(Your own installation path)
- Now you can connect the lab kit to computer via a "Type A to mini-B" USB cable. The lab kit should be correctly recognized as “STM32 STLink NODE L476RG”

Section 2: Install Tera Term

Since Nucleo-64 board does not have a LCD display, we will install a virtual terminal program, Tera Term, on our computer to display information. Please download the installer from https://ttssh2.osdn.jp/index.html.en and install the program. Tera Term installation package is already available and does not need to be downloaded from this link. teraterm-4.90

After installation, launch the program and select Serial Port: COM3:STMicroelectronics STLink Virtual COM Port and click OK. (Probably not COM3, you just focus "STMicroelectronics STLink.....")



Section 3: Project Creation

Summary

This tutorial takes the following the kit as an example of creating a project in Keil IDE for assembly programs.

- STM32L476RG MCU (Cortex-M4 with FPU and DSP)

Note that the project does not use the default startup files provided by Keil. You need to download a modified version of ***startup_stm32l476xx.s*** from Black board.

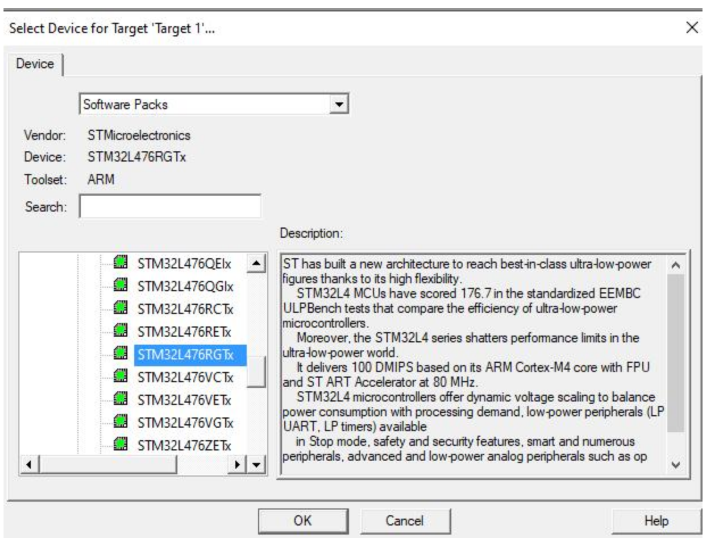
Identifying Target Processor

Kit	**Processor**	**Core**	**Flash**	**RAM**
STM32L476 Nucleo-64	**STM32L476RG**	**Cortex-M4 (DSP + FPU)**	**1 MB**	**128 KB**

Steps to create a new project in Keil

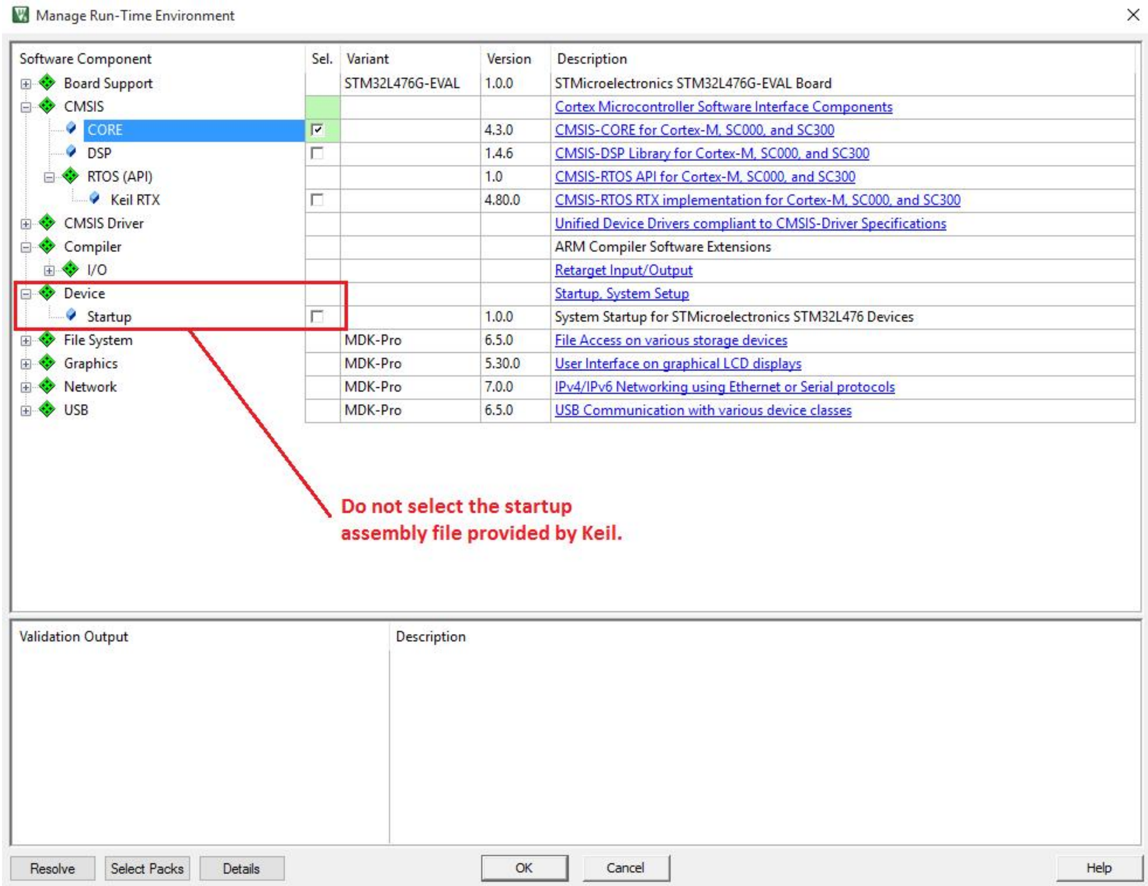
You'd better create a new folder firstly, to place your new projects in this folder.

1. From the menu **Project** ⟶ **New µVision Project**
2. Give the project a name and select its storage directory. In this tutorial, the project is named as "lab".
3. Select the device **STM32L4 Series**, and then select **STM32L476RGTx**.



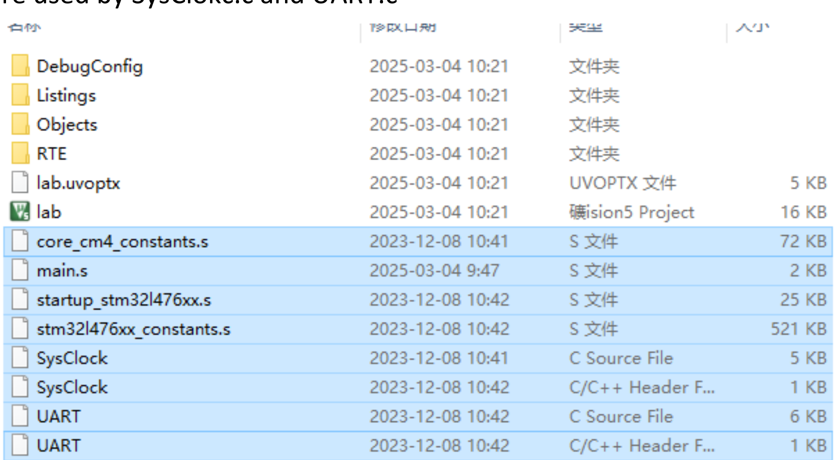
If did not see the targeted processor in the list, click the "**Pack Installer**" button from Menu Project -> Manage and install the component **Keil::STM32L4xx_DFP**.

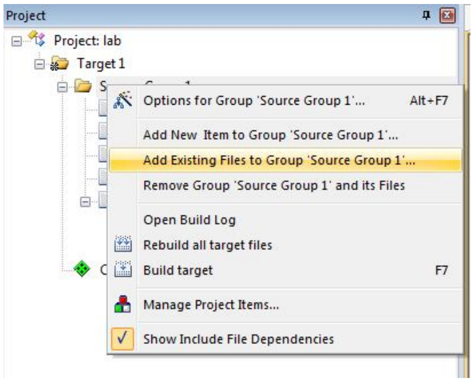
4. Select **CMSIS Core** only. Do NOT select "Device Startup". Instead, you should use the one provided by the course website.

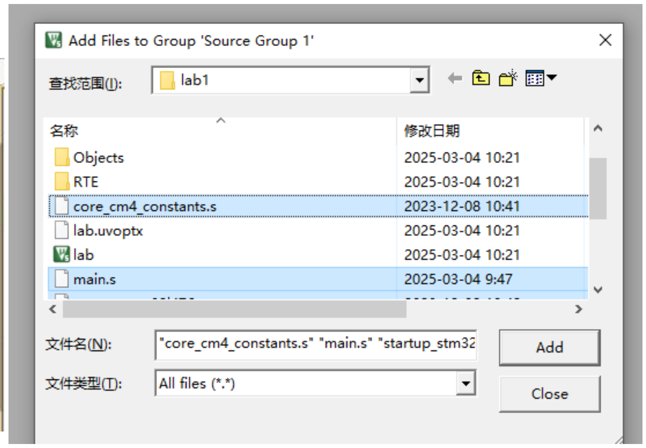


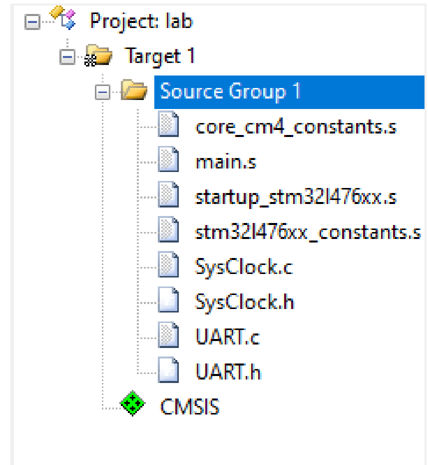
5. To create a project, you should download all the template files from Black board and put them in the project folder. Then right click on the folder “Target 1” -"Source Group 1"and add the following files as shown in the following figure:
    - **startup_stm32l476xx.s**
    - **main.s**
    - core_cm4_constrants.s
    - stm32l476xx_constants.s
    - SysClock.c
    - UART.c

Note that SysClock.h and UART.h are not added to the project, but these files must be in the project folder because they are used by SysClokc.c and UART.c



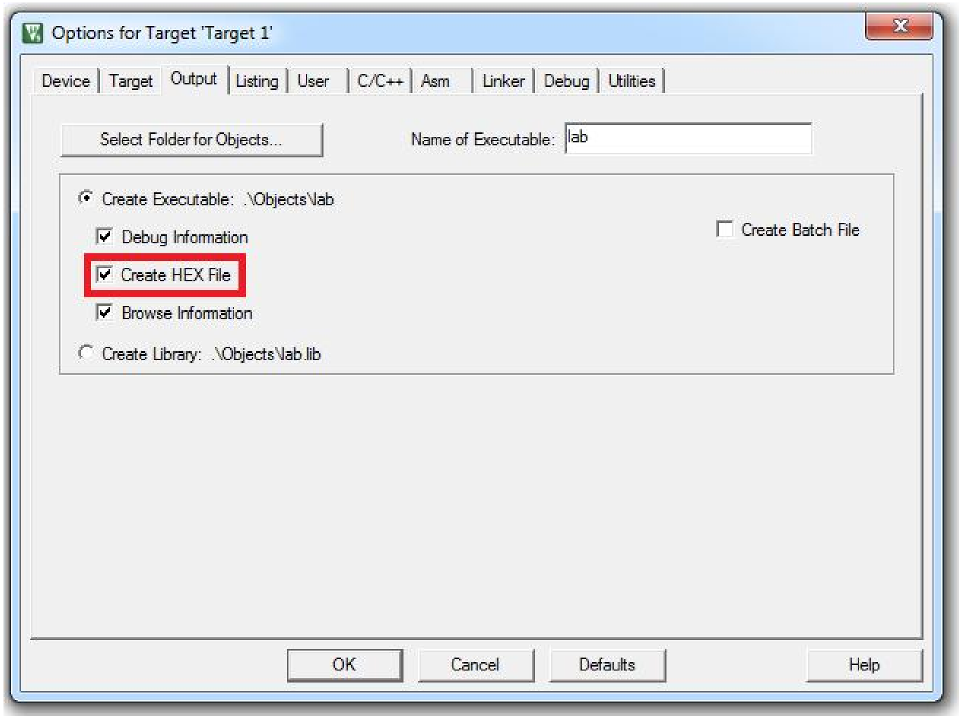






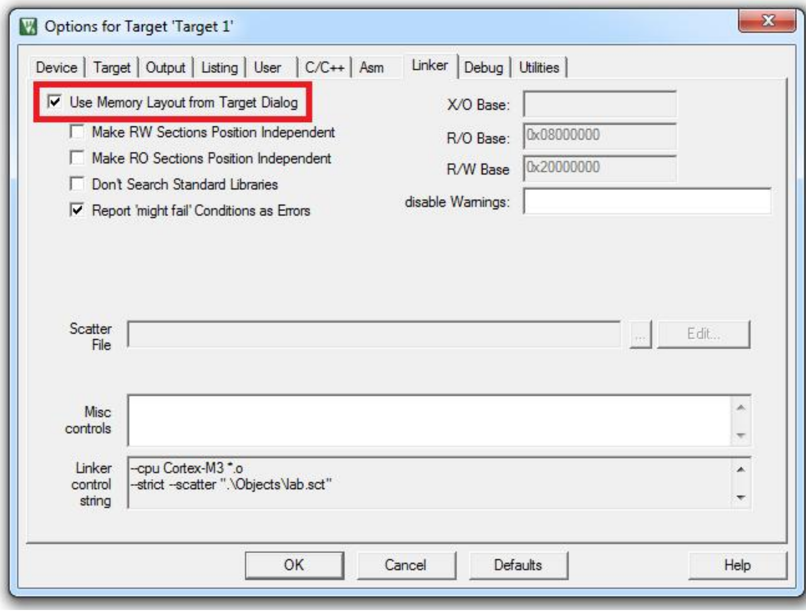
6. Set Project Properties

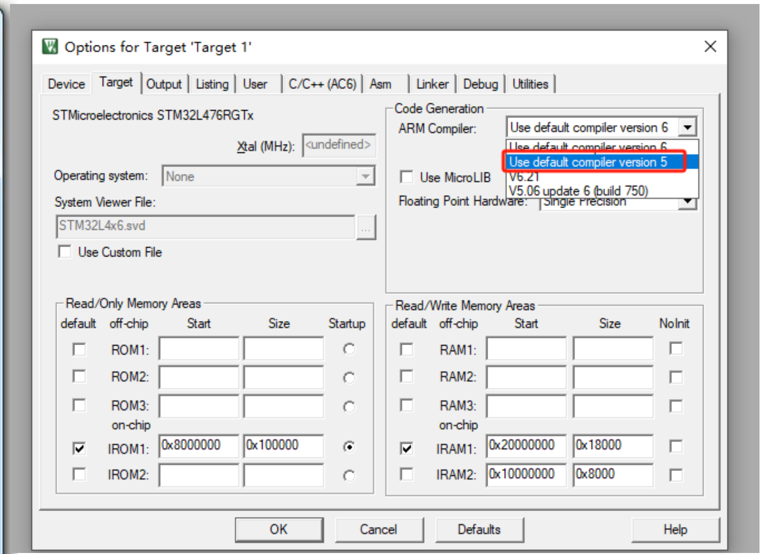
From the menu, click **Project** ⟶ **Option for Target**, Go to the **Output** page, select "**Create HEX file**"



Go to the **Linker** page, select "**Use Memory Layout from Target Dialog**"

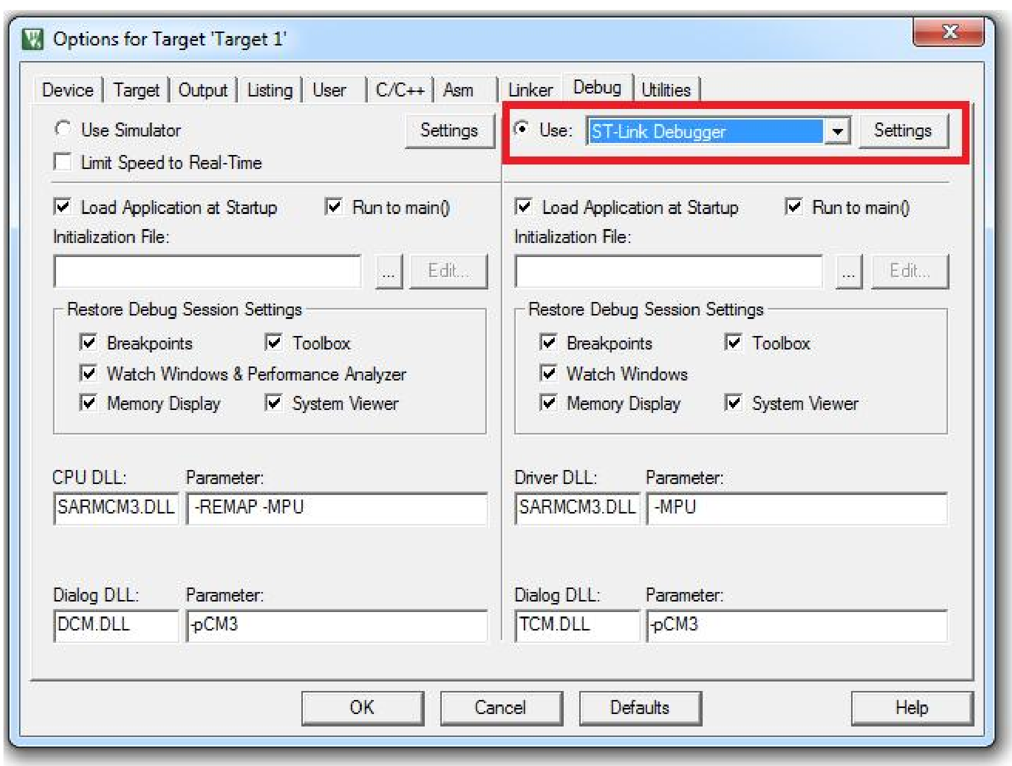
Go to the Target page, select "Use default compiler version 5"





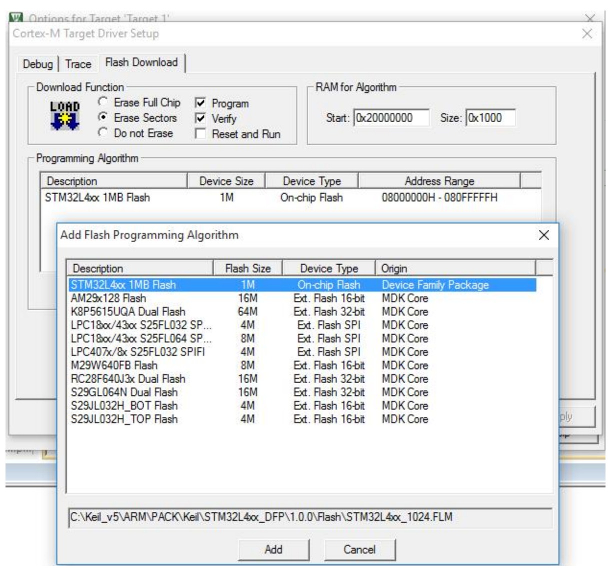
Go to the **Debug** page, select "**ST-Link Debugger**"

If you see a firmware upgrade notification, please follow the steps to upgrade the firmware.



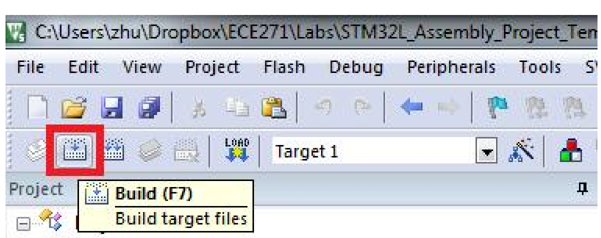
Click "**Settings**" and select "**SW**" (Serial Wire) as the port.

Go to the **Flash Download** page, and verity that **STM32L4xx On-chip Flash** is selected in the Programming Algorithm. If not, click "Add" and select STM32L4xx On-chip flash in the popped dialog.



7. Compile and run your project

Build the program:

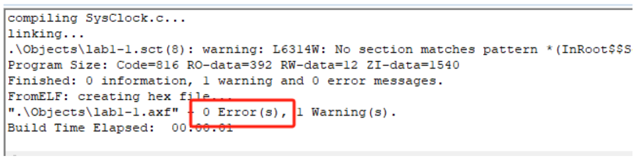


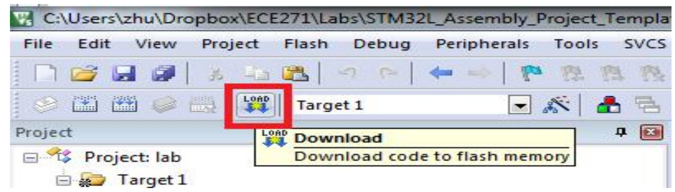
You can **ignore** the following warning when the linking stage:

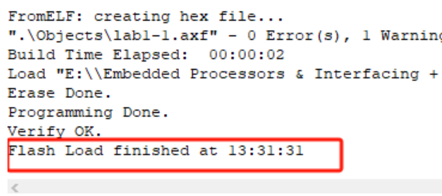
.\Objects\lab.sct(8): warning: **L6314W: No section matches pattern *(InRoot$$Sections)**.

Connect your lab kit to the computer and download the program to the STM32L processor.

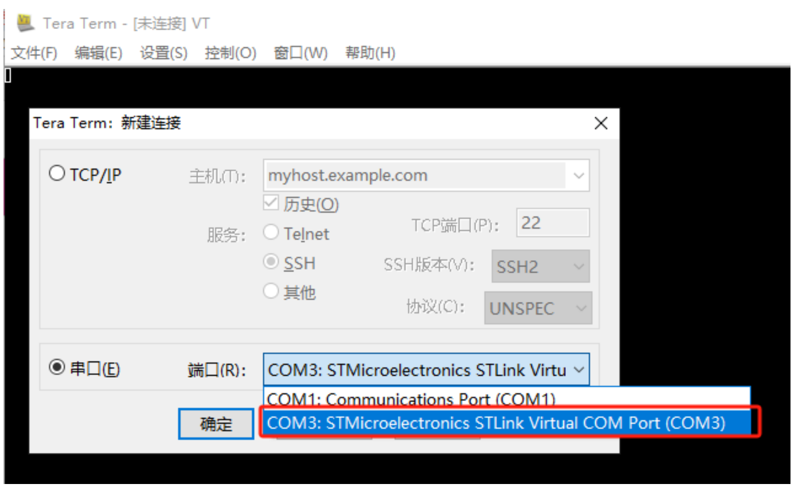
If there is any error, download cannot be successful, 0 error must

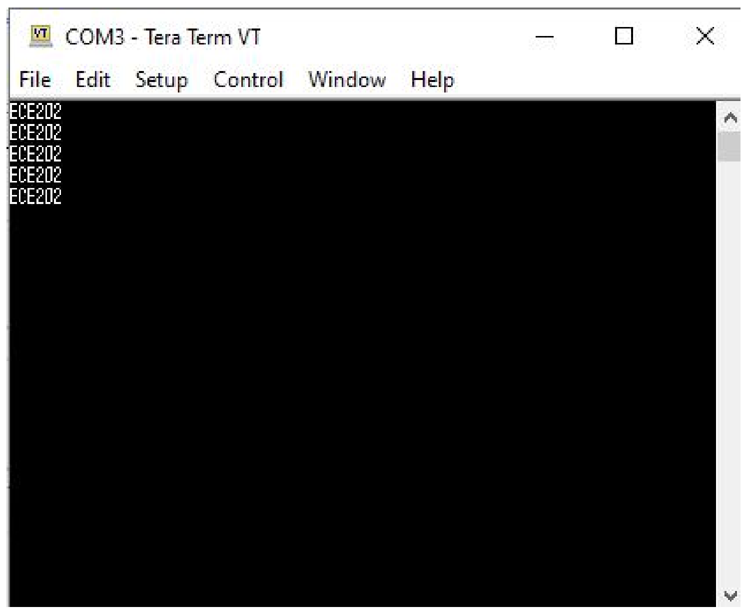






Push the black Reset button on the board, and your program will start. By default, ECE0202 will be displayed on the Tera Term each time you push the reset button.





Please replace the "ECE0202" in main.s with your last name, compile and download again. Your last name will be displayed on the LCD. Please do this for both team members.

Section 4: Debug

Follow the steps in course lecture. Take a screenshot of your last names saved into memory from the debugger (You may need to add memory address 0x20000000, Right-click and select ASCII)